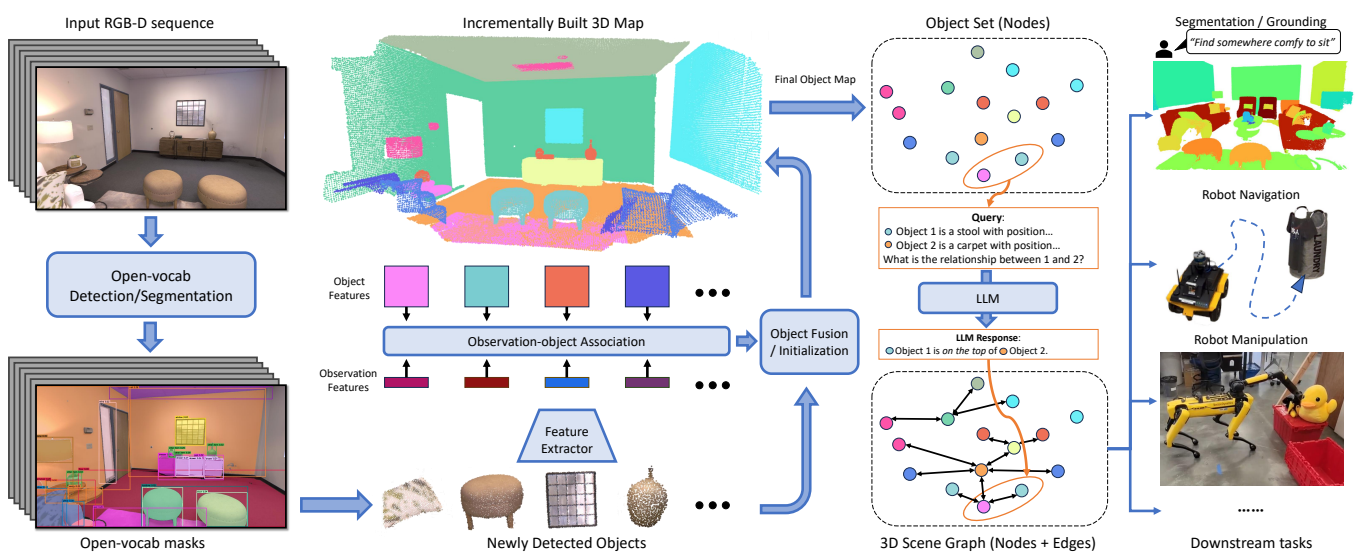

Input RGB-D sequence
Incrementally Built 3D Map
Object Set (Nodes)
Segmentation / Grounding
"Find somewhere comfy to sit"
Final Object Map
Open-vocab Detection/Segmentation
Object Features
Observation Features
Observation-object Association
Object Fusion / Initialization
Feature Extractor
Query:
Object 1 is a stool with position...
Object 2 is a carpet with position...
What is the relationship between 1 and 2?
LLM
LLM Response:
Object 1 is on the top of Object 2.
Robot Navigation
Robot Manipulation
Open-vocab masks
Newly Detected Objects
3D Scene Graph (Nodes + Edges)
Downstream tasks